UG (NEP) 5th SEMESTER
COMPUTER APPLICATIONS: OPERATING SYSTEM
CAP522J1

Date of Examination: 02-12-2025 No. of Questions: 09
Time Allowed: 2:00 Hours Maximum Marks: 75 Minimum Marks: 27

NOTE: Attempt all questions from Sections A and B and only two questions from Section C.

SECTION A: Short Answer Type Questions (8 × 1.5 = 12 Marks)

Answer all questions in a maximum of 20 words each.

1. What is an Operating System?
2. Define Process.
3. What is a binary semaphore?
4. What is a deadlock?
5. What is meant by a CPU-bound process?
6. Define inter-process communication.
7. What is external fragmentation?
8. List various page replacement algorithms.

SECTION B: Medium Answer Type Questions (4 × 7.5 = 30 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. Explain the operating system structure and its components.

OR

Consider the following processes and calculate average waiting time and turnaround time using SJF preemptive and SJF non-preemptive scheduling:

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9
P4	3	5

3. Define Process. Explain the Process State diagram.

OR

What is the critical section problem? Explain with an example.

4. What is a Semaphore? Explain the producer-consumer problem using semaphore.

OR

Discuss various deadlock prevention methods.

5. What is Segmentation? Explain with an example.

OR

Write short notes on: (a) Demand paging (b) Thrashing.

SECTION C: Long Answer Type Questions (2 × 16.5 = 33 Marks)

Answer only two questions in a maximum of 400 words each.

6. Explain the different functions of an operating system and discuss the various services provided by an operating system.
7. What is the difference between preemptive and non-preemptive scheduling algorithms? Explain FCFS scheduling algorithm. Find the average turnaround time and average waiting time for the processes given below by FCFS:

Process	CPU burst time (in ms)
P1	24
P2	3
P3	3

8. Explain the best fit, first fit and worst fit algorithms.
9. Consider the following page reference string:
   1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 2, 1, 2, 3, 6
   How many page faults would occur for the following replacement algorithms, assuming four frames?
   (i) LRU replacement
   (ii) FIFO replacement
   (iii) Optimal replacement